Final Project Report

Assessing the Impact of Data Cleaning on the Relationship Between Hospital Quality Ratings and Community Social Vulnerability

IS537 – Theory and Practice of Data Cleaning
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1. Introduction

Data integration projects often assume that datasets can be reliably combined using shared geographic or categorical identifiers. In practice, however, inconsistencies in formatting, naming conventions, typographical errors, and missing identifiers can significantly reduce integration quality and affect downstream analysis. These issues become especially important when combining datasets from different organizations that were collected for different purposes.

This project examines how data cleaning decisions affect the integration of hospital quality data with community-level social vulnerability data. The project combines the Centers for Medicare & Medicaid Services (CMS) Hospital General Information dataset with the Centers for Disease Control and Prevention (CDC) Social Vulnerability Index (SVI) dataset. The goal is to evaluate whether cleaning and standardizing geographic identifiers improves dataset integration quality and influences downstream analytical conclusions.

The central research question guiding this project is:

Does cleaning and standardizing geographic identifiers improve integration quality and affect downstream analysis outcomes when combining hospital quality data with county-level social vulnerability measures?

This question is strongly connected to the concept of fitness-for-use discussed throughout the course. The intended use case requires reliable integration between hospital-level and county-level datasets. If geographic identifiers are inconsistent or incomplete, the resulting dataset may not be fit for analysis because hospitals could fail to match to the correct county-level vulnerability measures.

Rather than treating cleaning as a preprocessing task performed independently from analysis, this project evaluates cleaning as an integral component of analytical reliability. The project specifically investigates what changed after cleaning, what did not change, and why.

2. Datasets

CMS Hospital General Information Dataset

The primary dataset used in this project is the CMS Hospital General Information dataset published through the CMS Provider Data catalog. The dataset contains hospital-level information including:

- Facility identifiers
- Hospital names
- Addresses and geographic information
- Hospital ownership types
- Emergency service availability
- Overall hospital quality ratings

The dataset contained approximately 5,426 hospital records.

A major limitation identified during profiling was that approximately 47% of hospitals did not contain an overall hospital rating and instead used the value "Not Available." This represented an important completeness issue because missing ratings reduce the amount of data available for downstream comparison.

CDC Social Vulnerability Index (SVI)

The second dataset used in this project was the CDC Social Vulnerability Index dataset. The SVI dataset contains county-level measures of social vulnerability derived from demographic and socioeconomic indicators.

The dataset includes measures related to:

- Socioeconomic status
- Household composition
- Minority status and language
- Housing and transportation

The project primarily used the overall SVI ranking variable (RPL_THEMES) as a summary measure of vulnerability.

The dataset contained approximately 3,144 county-level records.

U.S. Census County Reference File

A third supporting dataset was used during cleaning and integration. The U.S. Census county reference file provided standardized county-level Federal Information Processing Standard (FIPS) codes.

This dataset was used to map hospital county and state combinations to official FIPS codes and create a standardized integration key.

3. Use Case and Fitness-for-Use Reasoning

The primary analytical goal of the project was to evaluate whether hospitals located in socially vulnerable counties tend to receive lower hospital quality ratings.

To answer this question reliably, the integrated dataset needed to satisfy several quality requirements:

- Geographic identifiers needed to be consistent across datasets
- County-level matches needed to be accurate
- Integration coverage needed to be sufficiently high to avoid systematic exclusion of hospitals
- Missing values needed to be identified and assessed for their potential analytical impact

This made integration quality a critical component of fitness-for-use. If county identifiers were inconsistent or unmatched, hospitals could fail to receive the correct county-level SVI values, reducing both completeness and analytical reliability.

Join rate was therefore used as a primary quality metric because successful linkage between hospital records and county-level SVI measures directly determined whether the dataset was usable for downstream analysis.

4. Initial Data Profiling and Quality Assessment

Initial profiling was conducted to understand dataset structure, identify potential quality issues, and evaluate integration readiness.

Dataset Structure

Initial inspection identified:

Dataset	Records	Key Geographic Fields
CMS Hospital General Information	5,426	County/Parish, State
CDC SVI	3,144	COUNTY, ST_ABBR, FIPS

A major issue identified during profiling was that the CMS dataset did not contain FIPS codes, while the SVI dataset relied on FIPS as the primary geographic identifier.

Completeness Assessment

Completeness was assessed by evaluating missing values in critical fields.

The hospital rating field contained a substantial amount of missing data.

Metric	Value
Hospitals with missing ratings	~47%

This issue persisted before and after cleaning:

Dataset Version	Missing Rating Proportion
Before FIPS integration	0.4718
After FIPS integration	0.4702

This demonstrates that cleaning improved integration quality but did not resolve missingness inherent to the original hospital dataset.

Consistency Issues

Multiple inconsistencies were identified in county names and geographic identifiers.

Examples included:

Example Type	Example
Formatting inconsistency	“DE KALB” vs “DEKALB”
Abbreviation inconsistency	“E. BATON ROUGE”

Example Type	Example
Geographic suffix variation	“County” vs “Parish” vs “Borough”
Typographical error	“JEFFRSON” instead of “JEFFERSON”

These inconsistencies prevented reliable integration using direct string-based joins.

Validity Issues

Several county names contained spelling or formatting errors that reduced integration validity.

Examples included:

- “JEFFRSON DAVIS”
- “YELLOW MEDCINE”
- “BOROUH” instead of “BOROUGH”

These issues required targeted correction strategies during cleaning.

5. Initial Integration Attempt

An initial integration attempt was performed using standardized county and state string fields.

The join matched records based on cleaned county and state names.

Integration Method	Join Rate
Standardized string-based join	93.3%

This initial result demonstrated that direct matching based on geographic names alone produced incomplete integration coverage.

Although the result was relatively high, approximately 365 hospital records still failed to match correctly.

This indicated that additional cleaning and standardization steps were necessary.

6. Data Cleaning and Repair Workflow

To improve integration quality, multiple cleaning and repair methods were implemented.

These methods were selected to address specific quality issues identified during profiling. Standardization and normalization improved consistency in join keys, while targeted corrections addressed dataset-specific inconsistencies that could not be resolved through general rules.

Step 1: Standardization of Geographic Fields

County names were standardized using several normalization procedures:

- Conversion to uppercase text
- Removal of punctuation
- Removal of extra spacing
- Removal of geographic suffixes such as:
    - County
    - Parish
    - Borough
    - Census Area

This reduced inconsistencies caused by formatting variation.

Step 2: Typographical Error Correction

Targeted corrections were applied to county names containing spelling or formatting errors.

Examples included:

Incorrect Value	Corrected Value
JEFFRSON	JEFFERSON
YELLOW MEDCINE	YELLOW MEDICINE
BOROUH	BOROUGH

Step 3: Handling Abbreviations and Naming Variations

Additional rules were applied to normalize abbreviations and inconsistent geographic naming patterns.

Examples included:

Original Value	Standardized Value
E. BATON ROUGE	EAST BATON ROUGE
ST. JOHN BAPTIST	SAINT JOHN THE BAPTIST

Step 4: FIPS-Based Integration

The most important cleaning step involved mapping county and state combinations to official FIPS codes using the U.S. Census county reference file.

This step introduced a standardized geographic identifier shared across datasets.

The implementation phase reinforced the importance of FIPS-based integration after observing limitations in direct string-based joins.

After applying FIPS mapping:

Metric	Value
Hospitals successfully mapped to FIPS	99.9%

This represented a substantial improvement in integration readiness.

Step 5: Edge Case Handling

Several edge cases required additional targeted handling.

These included:

- Alaska borough naming structures
- Washington D.C. geographic conventions
- U.S. territory naming differences
- historical county naming changes

Examples included:

- Oglala Lakota County
- Alaska borough classifications
- U.S. Virgin Islands geographic structures

Some unmatched records remained intentionally unresolved to avoid introducing subjective or potentially incorrect mappings.

7. Final Integration Results

After applying cleaning and FIPS-based integration methods, the final join rate improved substantially.

Metric	Before FIPS Integration	After FIPS Integration
Join Rate	93.3%	98.0%
Matched Hospital Records	5,061	5,371

Cleaning and FIPS-based integration increased the number of matched hospital records by more than 300 hospitals.

This demonstrates that cleaning significantly improved dataset integration quality and increased the usable coverage of the dataset.

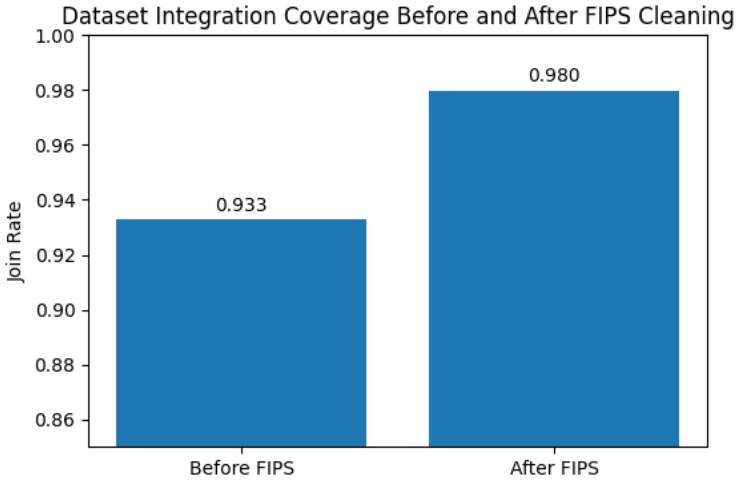

Figure 1. Dataset Integration Coverage Before and After FIPS Cleaning

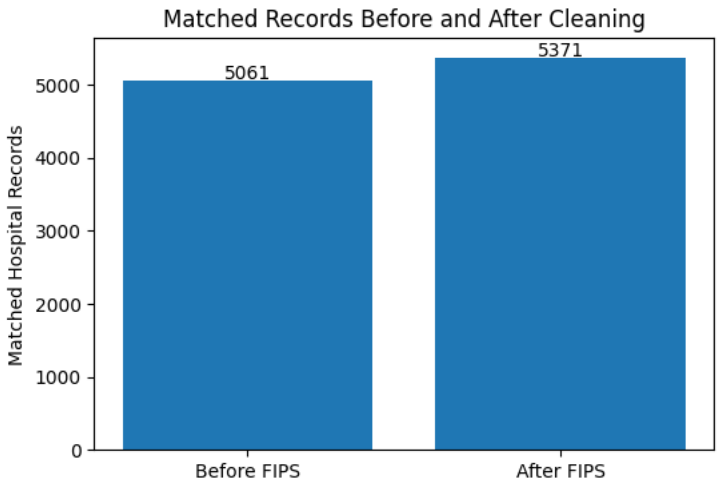

Figure 2. Matched Records Before and After Cleaning

8. Impact Assessment

The final stage of the project evaluated how cleaning affected downstream analytical outcomes.

The analysis examined the relationship between hospital quality ratings and county-level social vulnerability.

Before FIPS-Based Integration

Before FIPS-based integration, the average SVI by hospital rating was:

Hospital Rating	Average SVI
1	0.7379
2	0.6420
3	0.5701
4	0.5088
5	0.5021

The correlation between hospital rating and SVI was:

Metric	Value
Correlation (Rating vs SVI)	-0.2559

After FIPS-Based Integration

After FIPS-based integration, the average SVI by hospital rating was:

Hospital Rating	Average SVI
1	0.7340
2	0.6465
3	0.5783
4	0.5128
5	0.5011

The correlation between hospital rating and SVI was:

Metric	Value
Correlation (Rating vs SVI)	-0.2559

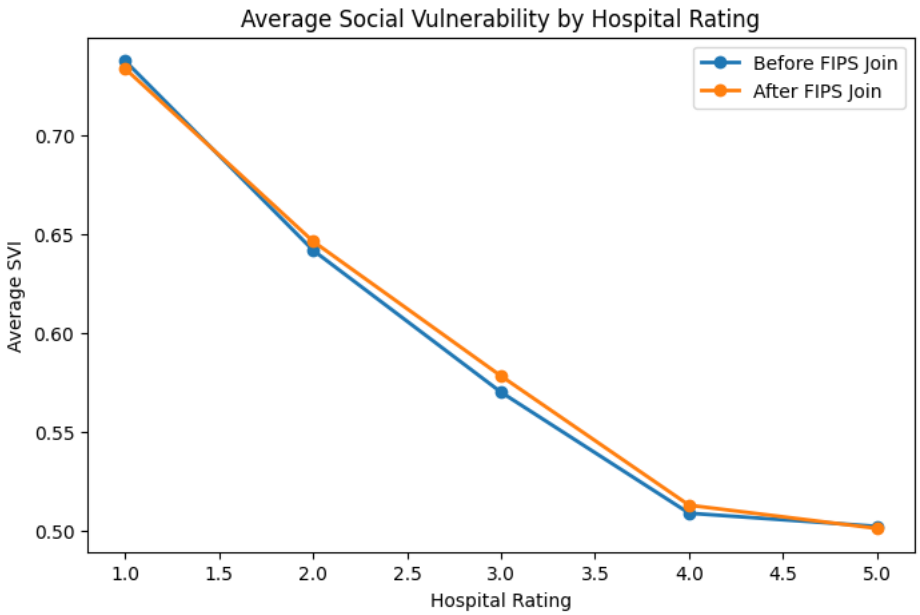

Figure 3. Average Social Vulnerability by Hospital Rating Before and After Cleaning

Interpretation of Results

Cleaning and FIPS-based integration substantially improved dataset coverage and increased the number of matched hospital records. This demonstrates that standardization and structured geographic mapping significantly improved integration quality.

However, despite the increase in coverage, the overall relationship between hospital ratings and social vulnerability remained largely unchanged. The correlation between hospital rating and SVI was nearly identical before and after FIPS-based integration.

Similarly, the pattern of lower-rated hospitals being associated with higher social vulnerability remained stable across both versions of the dataset.

These findings suggest that the primary impact of cleaning was improving completeness, confidence, and integration reliability rather than fundamentally altering the observed analytical relationship.

This distinction is important because it demonstrates that cleaning can improve dataset fitness-for-use even when broad analytical patterns remain stable.

The results also suggest that the relationship between hospital quality and social vulnerability was relatively robust despite initial integration inconsistencies.

9. Remaining Limitations

Several limitations remained after cleaning.

Missing Hospital Ratings

Approximately 47% of hospitals did not contain overall quality ratings.

This limited the completeness of downstream analysis and may introduce bias into the interpretation of hospital quality patterns.

Geographic Edge Cases

Some unmatched records remained due to structural geographic differences rather than correctable errors.

Examples included:

- Alaska borough structures
- U.S. territories
- special geographic naming conventions

These records were not forcefully matched in order to preserve reproducibility and avoid introducing subjective mappings.

Temporal Mismatch

Another limitation is the temporal mismatch between datasets.

The CMS hospital dataset reflects 2026 hospital information, while the SVI dataset represents 2022 vulnerability measures.

This temporal difference may affect alignment between hospital performance and county-level social vulnerability conditions.

10. Reflection and Lessons Learned

One important lesson from this project was that improving integration quality does not necessarily change analytical conclusions.

While cleaning significantly improved integration coverage and reduced mismatched records, the overall relationship between hospital ratings and social vulnerability remained stable.

This highlighted the importance of distinguishing between improving dataset reliability and changing analytical outcomes.

The project also demonstrated that data cleaning is not simply a preprocessing task. Cleaning decisions directly affect dataset usability, integration reliability, and confidence in downstream analysis.

Additionally, the project reinforced the importance of using standardized identifiers such as FIPS codes when integrating multi-source geographic datasets.

11. Reproducibility and Workflow Transparency

The project workflow was designed to be reproducible and transparent.

Artifacts included:

- Raw datasets
- Processed datasets
- Intermediate integration outputs
- Jupyter notebooks documenting cleaning workflows
- Summary metrics
- Visualizations used for evaluation

The repository structure separated raw data, processed outputs, notebooks, and documentation to support reproducibility.

The full workflow followed a consistent sequence:

1. Data loading and profiling
2. Initial integration attempt
3. Quality assessment
4. Cleaning and repair
5. FIPS mapping
6. Final integration
7. Impact evaluation

Github Link: https://github.com/kritikasingh62/IS537-Data-Cleaning-Project/tree/main

12. Conclusion

This project evaluated how data cleaning affected integration quality and downstream analysis when combining hospital quality data with social vulnerability measures.

The project demonstrated that cleaning and standardization significantly improved integration coverage, increasing join quality from approximately 93% to 98% and adding more than 300 matched hospital records.

At the same time, the overall relationship between hospital quality ratings and social vulnerability remained stable before and after cleaning.

These findings suggest that cleaning primarily improved completeness, integration reliability, and confidence in the dataset rather than fundamentally altering analytical conclusions.

Overall, the project demonstrates that data cleaning is a critical component of analytical reliability and fitness-for-use, particularly in multi-source integration workflows where inconsistent identifiers can substantially affect dataset usability.

13. References

1. Centers for Disease Control and Prevention, Agency for Toxic Substances and Disease Registry. (2022). *CDC Social Vulnerability Index 2022 database.* https://www.atsdr.cdc.gov/place-health/php/svi/svi-data-documentation-download.html?CDC_AAref_Val=https://www.atsdr.cdc.gov/placeandhealth/svi/data_documentation_download.html
2. Centers for Medicare & Medicaid Services. (2026). *Hospital General Information dataset.* https://data.cms.gov/provider-data/dataset/xubh-q36u
3. U.S. Census Bureau. (2023). *National county codes file.* https://www2.census.gov/geo/docs/reference/codes/files/national_county.txt
4. University of Illinois Urbana-Champaign. (2026). *IS537: Theory and Practice of Data Cleaning course materials and lecture notes.* School of Information Sciences.
5. Use of Chat GPT to assist in code generation and format the structure of this document